

Answer Key — Mathematics Grade 10

Sub-Strand 3.1: Trigonometry I • Lesson 4: Tangent and the Sine-Cosine-Tangent Relationship: Solving the Mango-Tree Problem

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5
C	B	B	B	B

USE AFTER: OBSERVE PHASE — Part A: algebraic derivation of $\tan \theta$

Q1. In a right-angled triangle, which ratio equals $\tan \theta$?

- A. Opposite / Hypotenuse
- B. Adjacent / Hypotenuse
- C. Opposite / Adjacent**
- D. Hypotenuse / Opposite

Correct answer: C. $\tan \theta = O/A$. O/H is $\sin \theta$ and A/H is $\cos \theta$. It follows from $\sin \theta \div \cos \theta$ with H cancelling.

USE AFTER: OBSERVE PHASE — Part B: numerical verification with mathematical tables

Q2. $\sin 60^\circ = 0.8660$ and $\cos 60^\circ = 0.5000$. Use $\tan \theta = \sin \theta / \cos \theta$ to find $\tan 60^\circ$.

- A. 0.5774
- B. 1.7320**
- C. 0.4330
- D. 1.3660

Correct answer: B. $0.8660 \div 0.5000 = 1.7320$, which matches $\tan 60^\circ$ in the tables (1.7321 to four decimal places). 0.5774 is the reciprocal ($\cos \div \sin$); 0.4330 is the product.

USE AFTER: EXPLAIN PHASE — Discussion Q2: which side is the 9.0 m shadow?

Q3. The sun makes a 53° angle with the ground and a tree casts a 9.0 m shadow. Relative to the 53° angle, the shadow is the triangle's ...

- A. Opposite side
- B. Adjacent side**
- C. Hypotenuse
- D. It depends on how tall the tree is

Correct answer: B. The shadow lies next to the 53° angle and is not the hypotenuse, so it is the adjacent side. The tree height is opposite; the sun ray is the hypotenuse.

USE AFTER: EXPLAIN PHASE — Part B: mango-tree height calculation

Q4. $\tan 53^\circ = h / 9.0$. Which step correctly finds h ?

- A. $h = 9.0 \div \tan 53^\circ$
- B. $h = 9.0 \times \tan 53^\circ$**
- C. $h = \tan 53^\circ \div 9.0$
- D. $h = 9.0 + \tan 53^\circ$

Correct answer: B. Multiply both sides by 9.0 to get $h = 9.0 \times \tan 53^\circ$. The lesson plan flags $h = 9.0 \div \tan 53^\circ$ as the common error.

USE AFTER: EXPLAIN PHASE — After the mango-tree calculation (new application problem)

Q5. A flagpole casts a 10 m shadow when the sun angle is 40° . $\tan 40^\circ = 0.8391$. How tall is the flagpole?

- A. 0.84 m
- B. 8.39 m**
- C. 10.84 m
- D. 11.92 m

Correct answer: B. $h = 10 \times \tan 40^\circ = 10 \times 0.8391 = 8.39$ m. 11.92 m comes from dividing ($10 \div 0.8391$), the same error as in Quick Check 4.

© 2026 SeaVuria and ARES. Aligned with KICD objectives. Developed by SeaVuria; technical implementation by ARES (with AI assistance from Claude). Licensed under CC BY-NC 4.0 (<https://creativecommons.org/licenses/by-nc/4.0/>).